Kommana Srinivasa Rao

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Summary

AI Data Science undergraduate (IIIT Sri City, CGPA 9.03) building production GenAI systems — hybrid RAG, multi-agent LLM orchestration, multilingual speech — with rigorous, measured evaluation.

Education

Indian Institute of Information Technology, Sri City Aug 2023 – Present
B.Tech in Artificial Intelligence and Data Science | CGPA: 9.03/10.0 *Sri City, AP*

Experience

Defence Research and Development Organisation (DRDO) May 2026 – July 2026
AI Engineering Intern

- Built a **4-agent LangGraph + Gemini system** that auto-diagnosed and repaired **70% of injected pipeline failures (7/10)** across **7 categories and 4 severity tiers**, healing schema drift in **under 3 seconds**.
- Engineered a **dry-run/rollback actuator** and **Pydantic v2 quarantine layer** that isolates bad rows instead of aborting runs – **90% throughput** on dirty batches, **9/12 runs with zero data loss**.
- Shipped an **11-endpoint FastAPI control plane**, live ops dashboard, and **dual SQLite/PostgreSQL adapter** for zero-change serverless deployment on Vercel.
- **Tech:** Python, LangGraph, Gemini, FastAPI, Pydantic v2, PostgreSQL, SQLite, Vercel.

Projects

Basha – Multilingual Text-to-Speech & Localization Service

- Built a **CPU-only FastAPI service** translating and narrating English into **11 languages**, with swappable TTS/translation backends and a multi-voice audio-drama mode casting a distinct voice per character.
- Evaluated on a **1,012-sentence FLORES-200 set** via ASR round-trip and multilingual-BERT similarity – **0.80–0.92 across all 11 languages at 100% success**, with **RTF ∼0.1 (10× real-time)** on CPU.
- Productionized with **SHA-256 audio caching**, async jobs, a **45-case pytest suite**, and **Dockerized Render deployment**.
- **Tech:** Python, FastAPI, Edge-TTS, gTTS, deep-translator, sentence-transformers, ffmpeg, Docker, pytest.

QdrantRERANK – Dual-Stage Hybrid RAG with Enforced Grounding

- Built a search + answer system over **93,455 StackExchange answers (218K chunks)**, combining keyword and semantic search with a **Cross-Encoder re-ranker**; reduced answer latency from **∼15s to 1–2s**.
- Enforced **code-level grounding gates** that block generation when retrieved evidence is insufficient, preventing unsupported answers instead of relying solely on LLM prompting.
- Built a shared **safety and validation layer** across the web UI and API, and fixed streaming failures including duplicated concatenated answers and stalled requests with **timeouts**.
- **Tech:** Python, FastAPI, Qdrant, Groq, LangChain, Gradio, sentence-transformers, Docker.

Technical Skills

Languages: Python,SQL
GenAI ML: LangGraph, LangChain, RAG, Hybrid Search Re-Ranking, Vector Databases (Qdrant, ChromaDB), Hugging Face / sentence-transformers, Google Gemini, Groq, DeepEval, TensorFlow, Scikit-learn
Backend Data: FastAPI, Pydantic v2, REST APIs, PostgreSQL, SQLite, ETL, NumPy, Pandas
Tools Deployment: Docker, Git, Vercel, Render, LangSmith, structlog, pytest, Jupyter